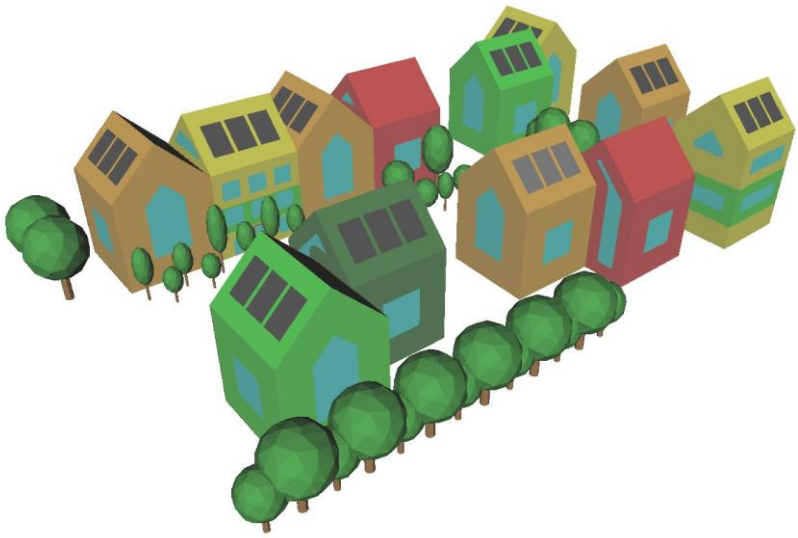
CityGML

energyADE 3.0

Beta 8

ER Mapping



Last updated on 16 October 2026

DigiTwins4PEDs

FlexPED

RENODAT

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The latest version of this document can be downloaded from:
https://github.com/tudelft3d/Energy_ADE/tree/3.0beta8/documentation

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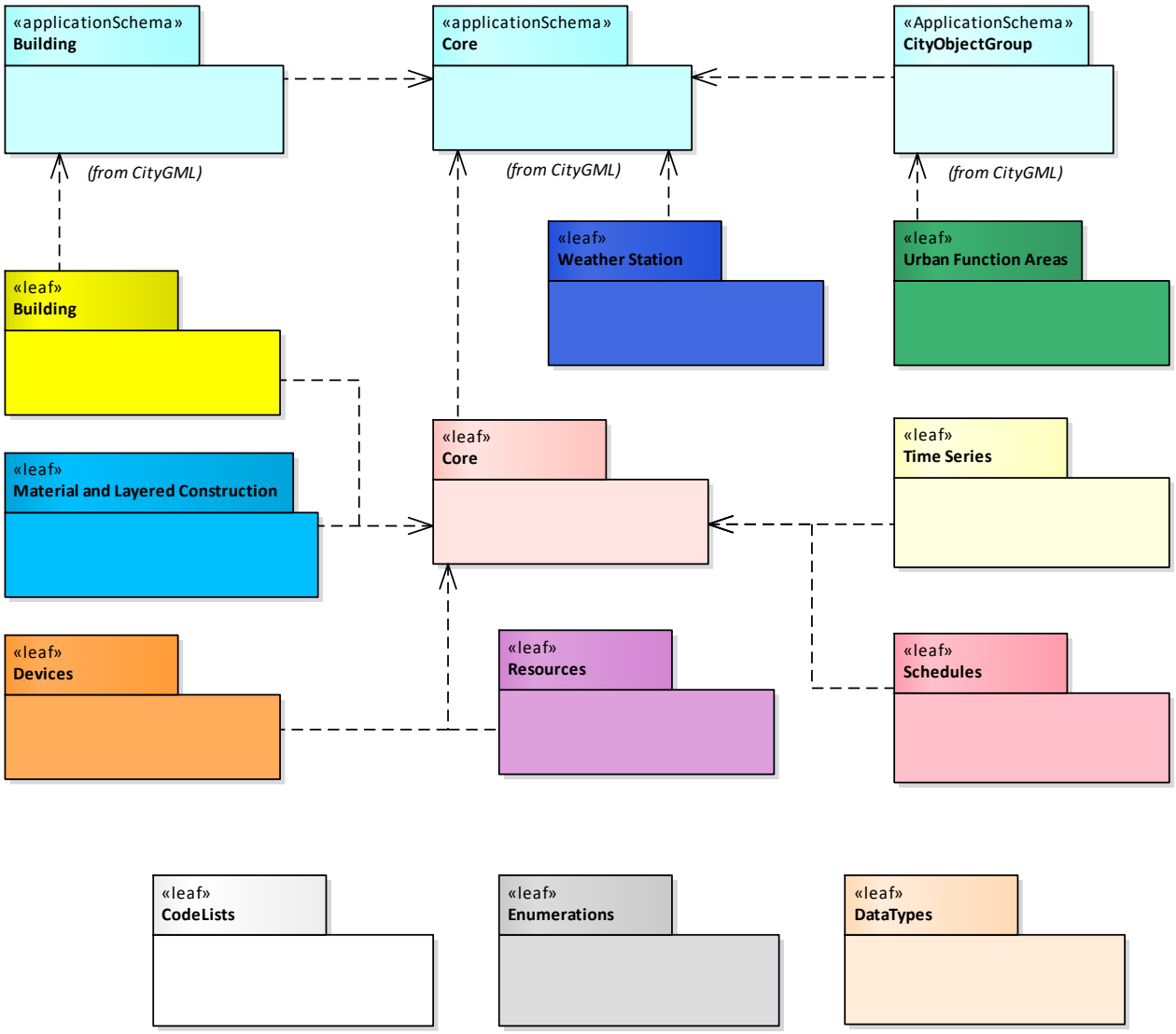
Since 2025, several other partners have joined on a volunteer basis as contributors, early testers or technology enthusiasts, who further develop the Energy ADE as a joint, open project.

Particular gratitude is due to Daniele Gasparini for designing the new Energy ADE logo.

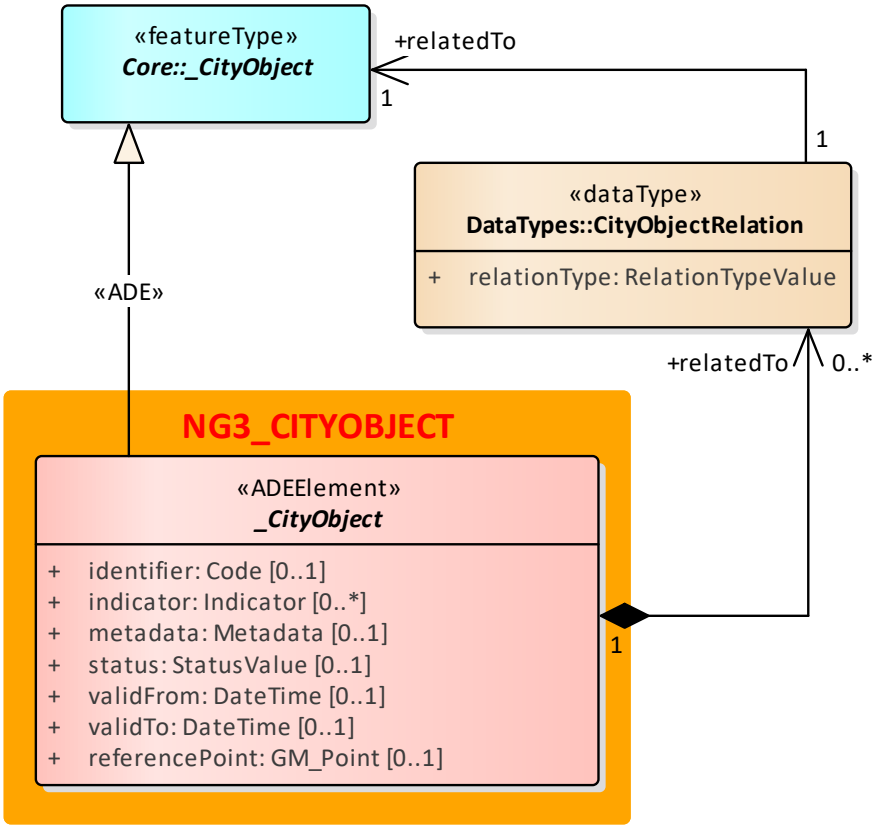
Contents

Overview of the packages 3
Core package part 1 4
Core package part 2 5
Core package part 3 7
Building package part 1 8
Building package part 2 9
Building package part 3 10
Devices package part 1 11
Devices package part 2 12
Urban function areas package 13
Weather station package 13
Resources package 14
Layered construction package 15
Schedules package 16
Time series package part 1 17
Time series package part 2 18
DataTypes package 19

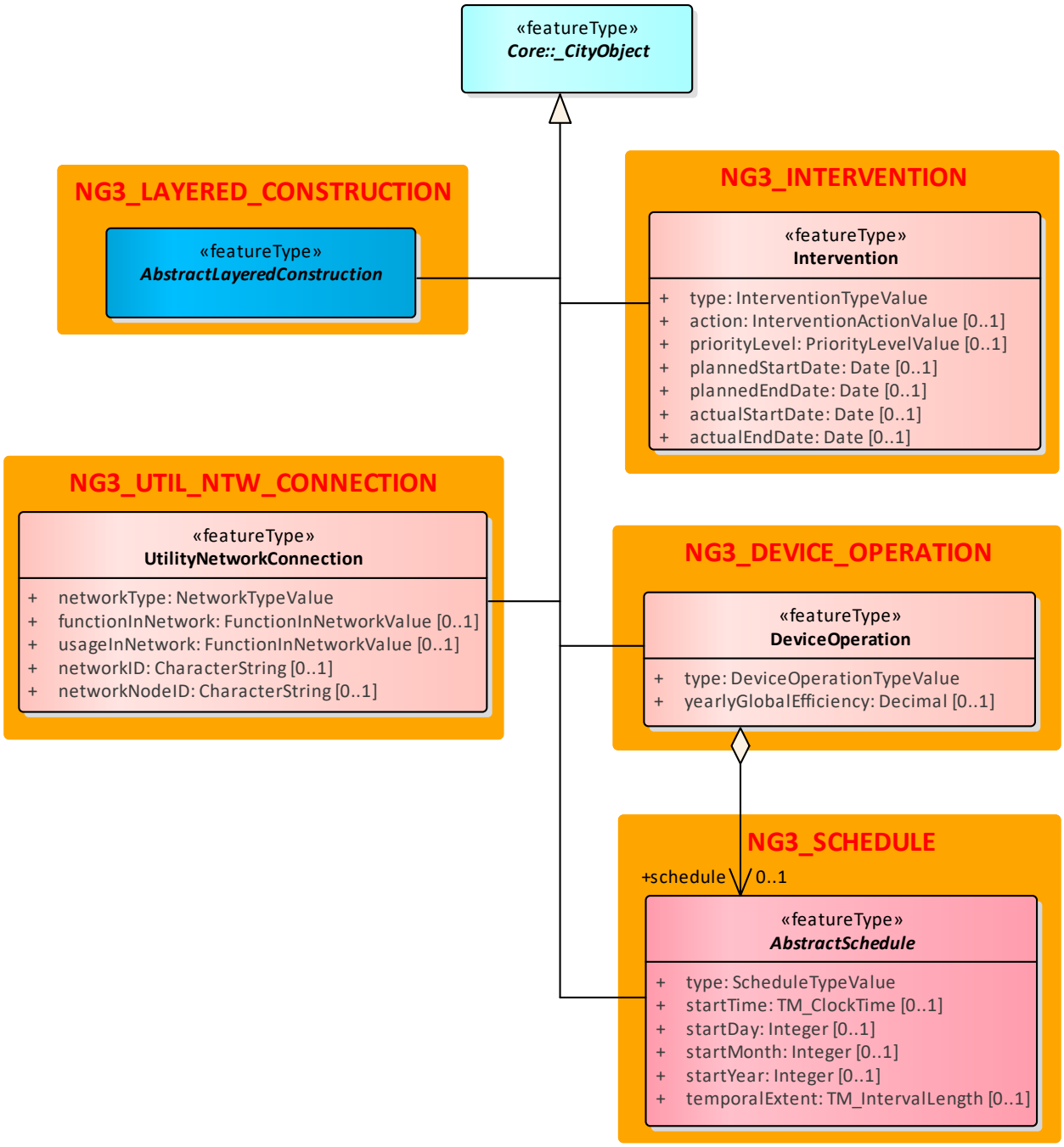
Overview of the packages

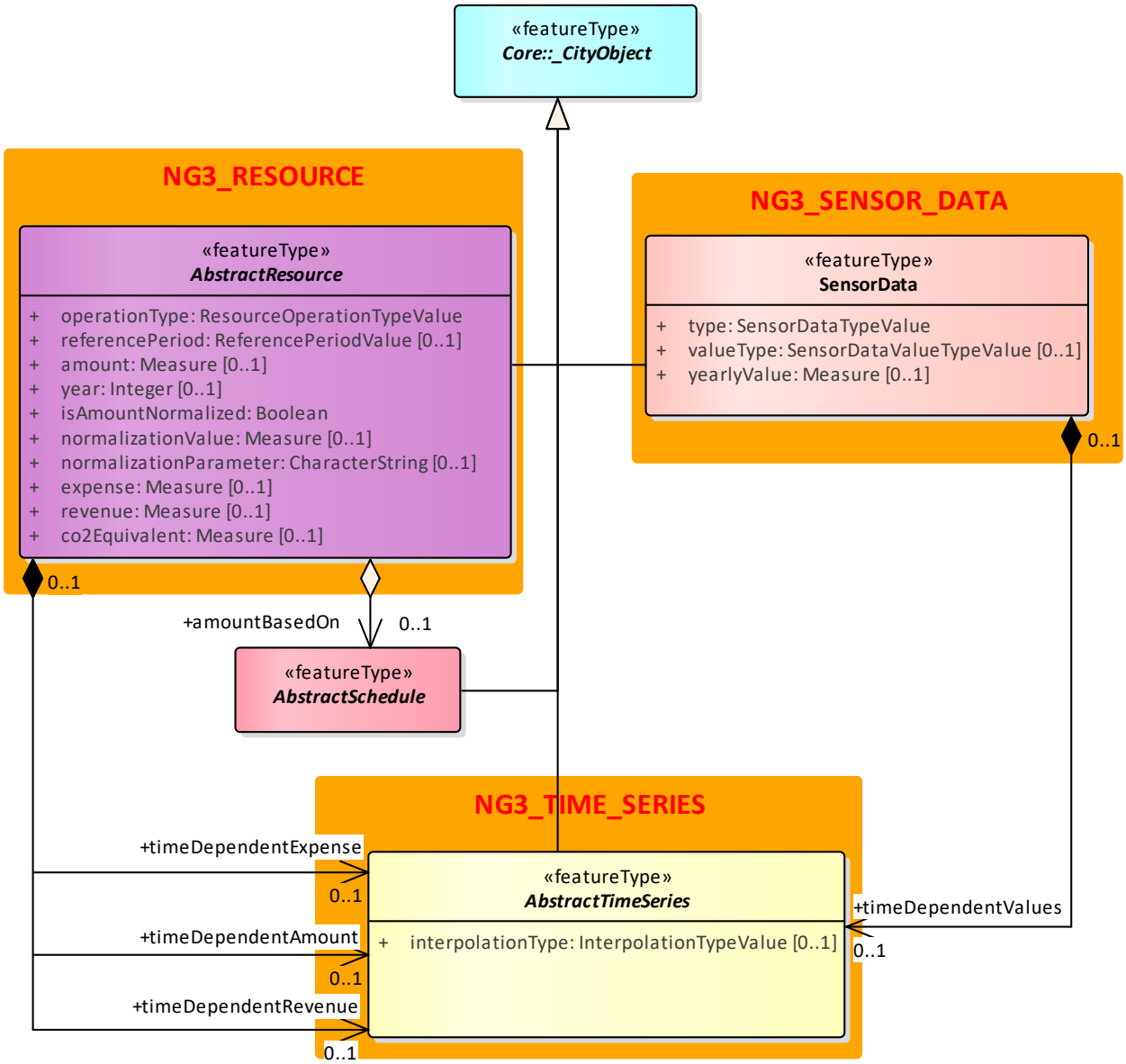


Core package part 1

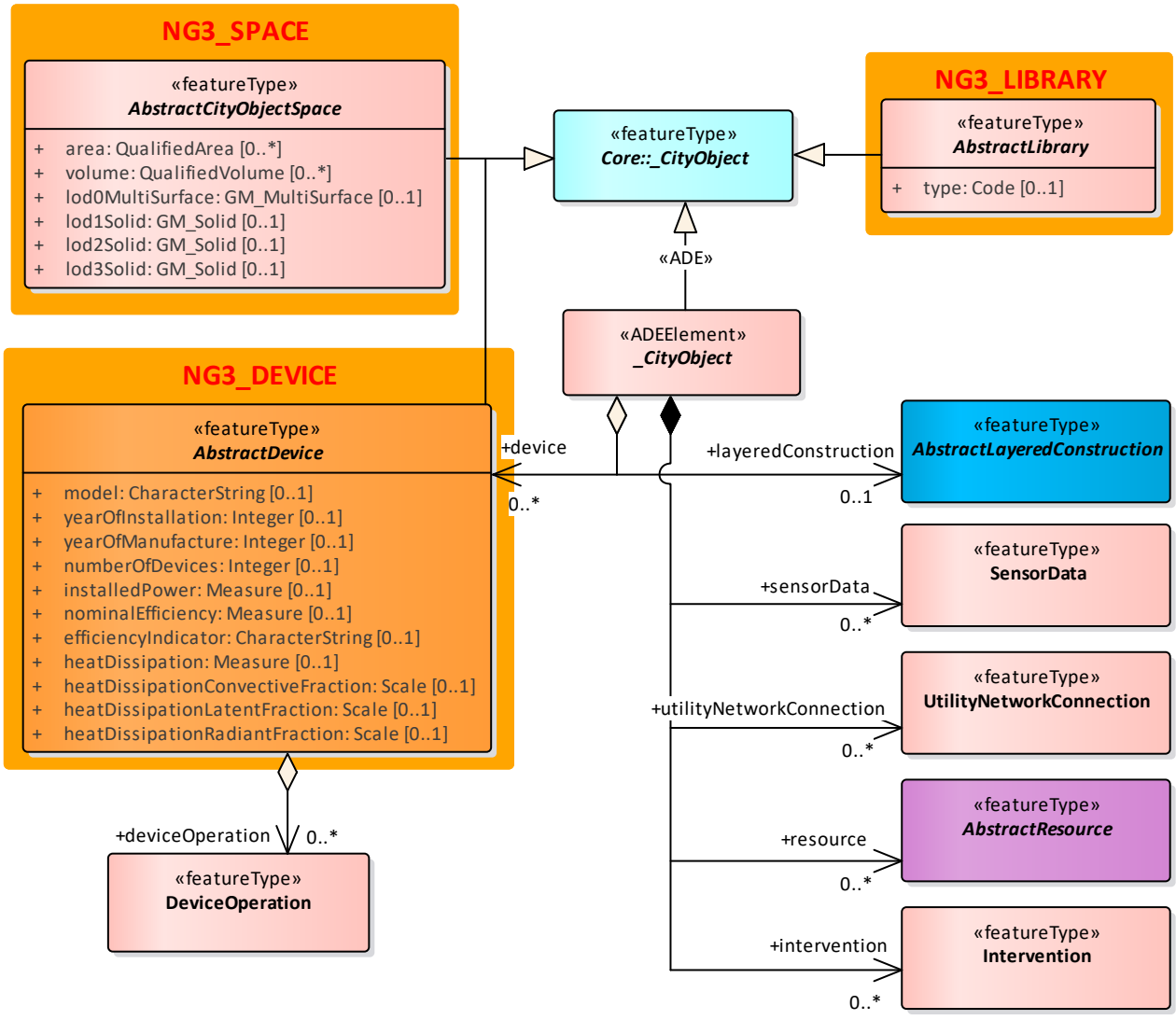


Core package part 2

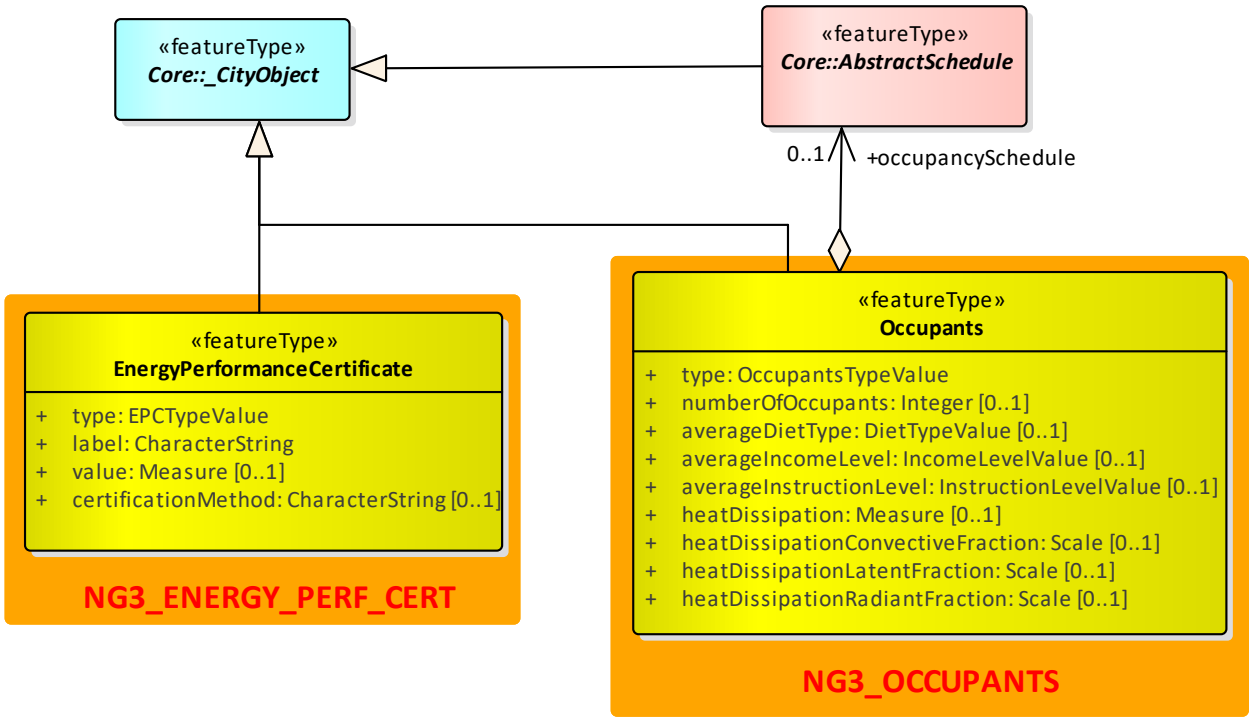



«featureType»
Core::_CityObject
NG3_RESOURCE
«featureType»
AbstractResource
+ operationType: ResourceOperationTypeValue
+ referencePeriod: ReferencePeriodValue [0..1]
+ amount: Measure [0..1]
+ year: Integer [0..1]
+ isAmountNormalized: Boolean
+ normalizationValue: Measure [0..1]
+ normalizationParameter: CharacterString [0..1]
+ expense: Measure [0..1]
+ revenue: Measure [0..1]
+ co2Equivalent: Measure [0..1]
NG3_SENSOR_DATA
«featureType»
SensorData
+ type: SensorDataTypeValue
+ valueType: SensorDataValueTypeValue [0..1]
+ yearlyValue: Measure [0..1]
0..1
0..1
+amountBasedOn
0..1
«featureType»
AbstractSchedule
NG3_TIME_SERIES
+timeDependentExpense
0..1
+timeDependentAmount
0..1
+timeDependentRevenue
0..1
«featureType»
AbstractTimeSeries
+ interpolationType: InterpolationTypeValue [0..1]
+timeDependentValues
0..1

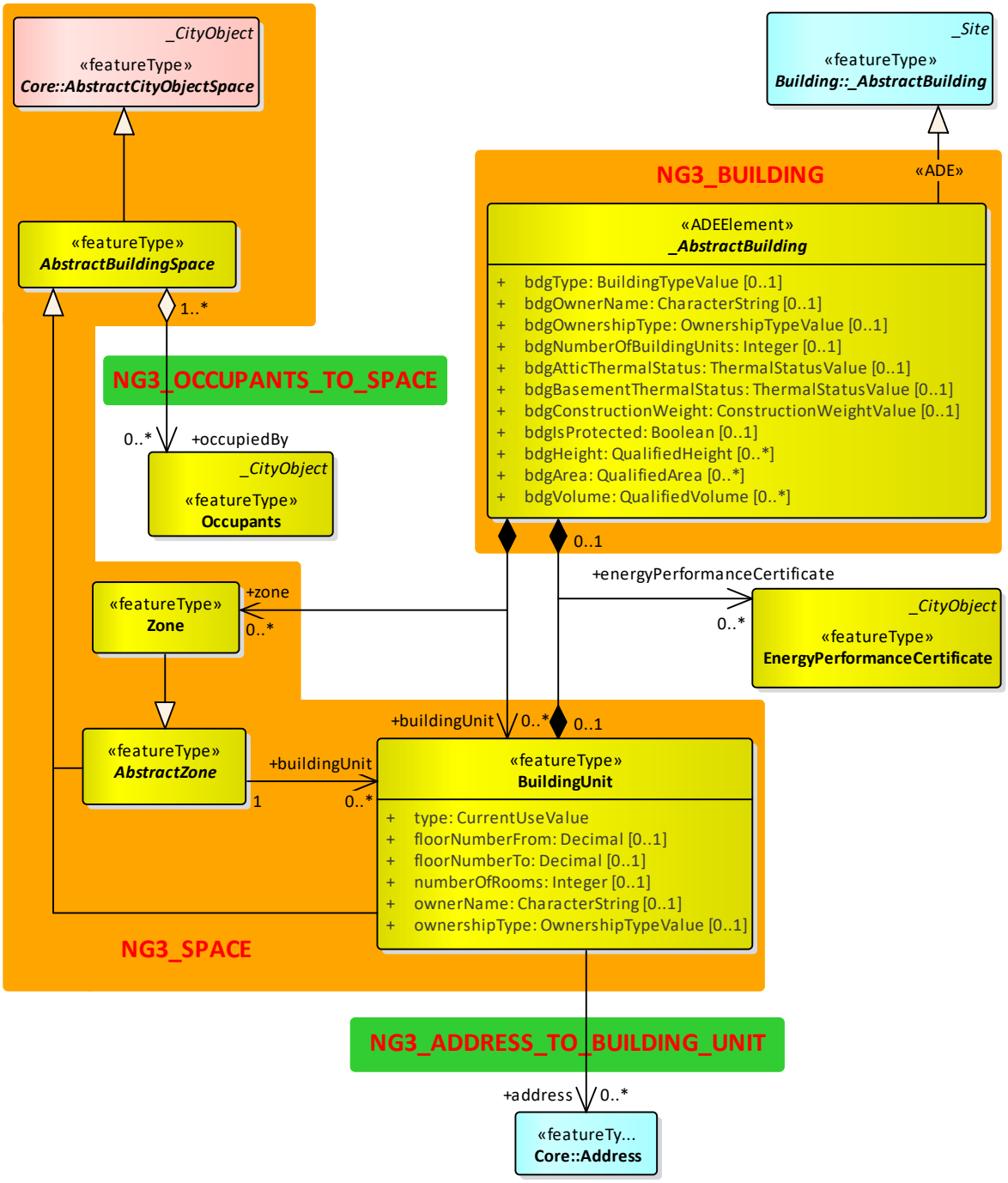
Core package part 3



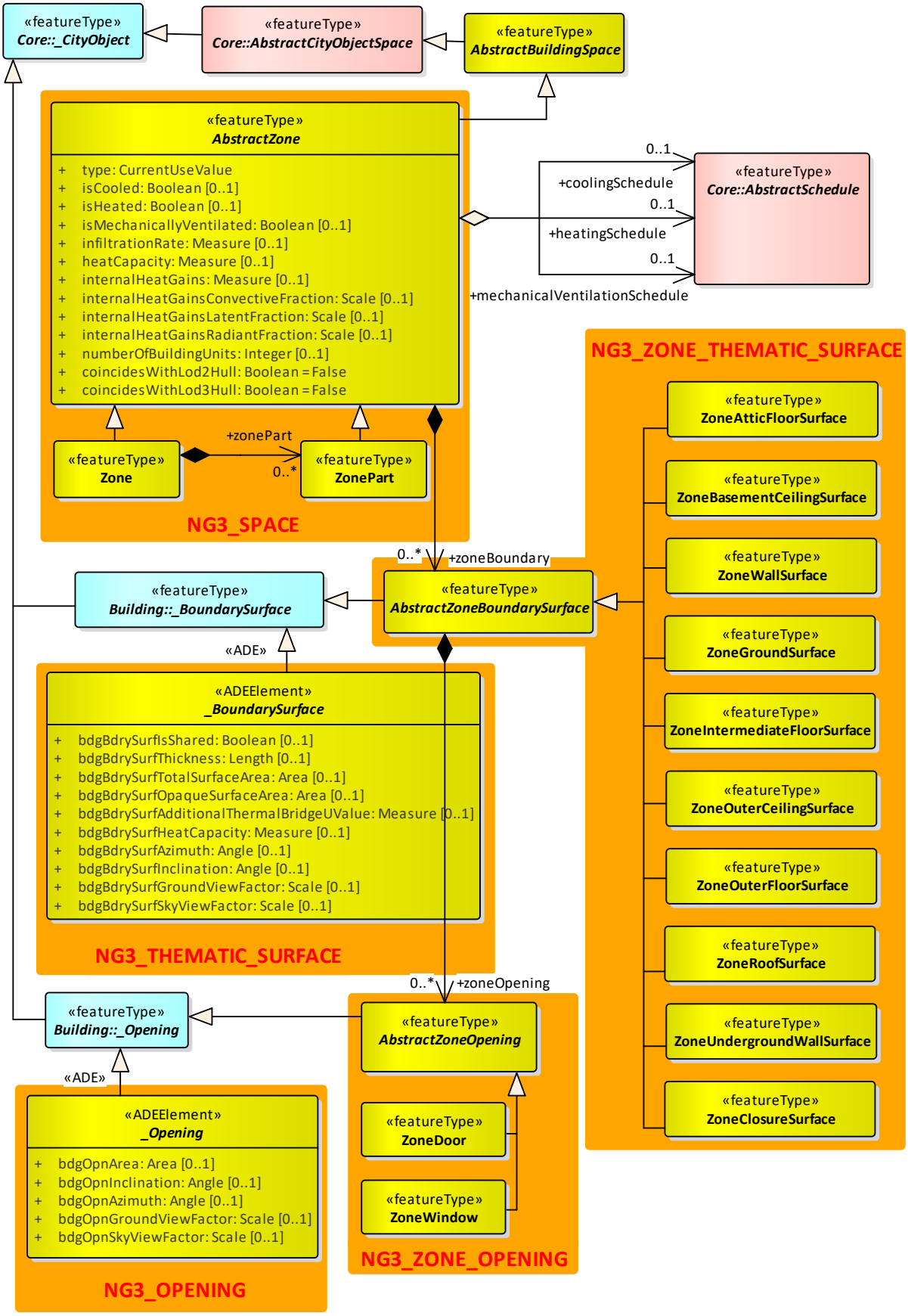
Building package part 1



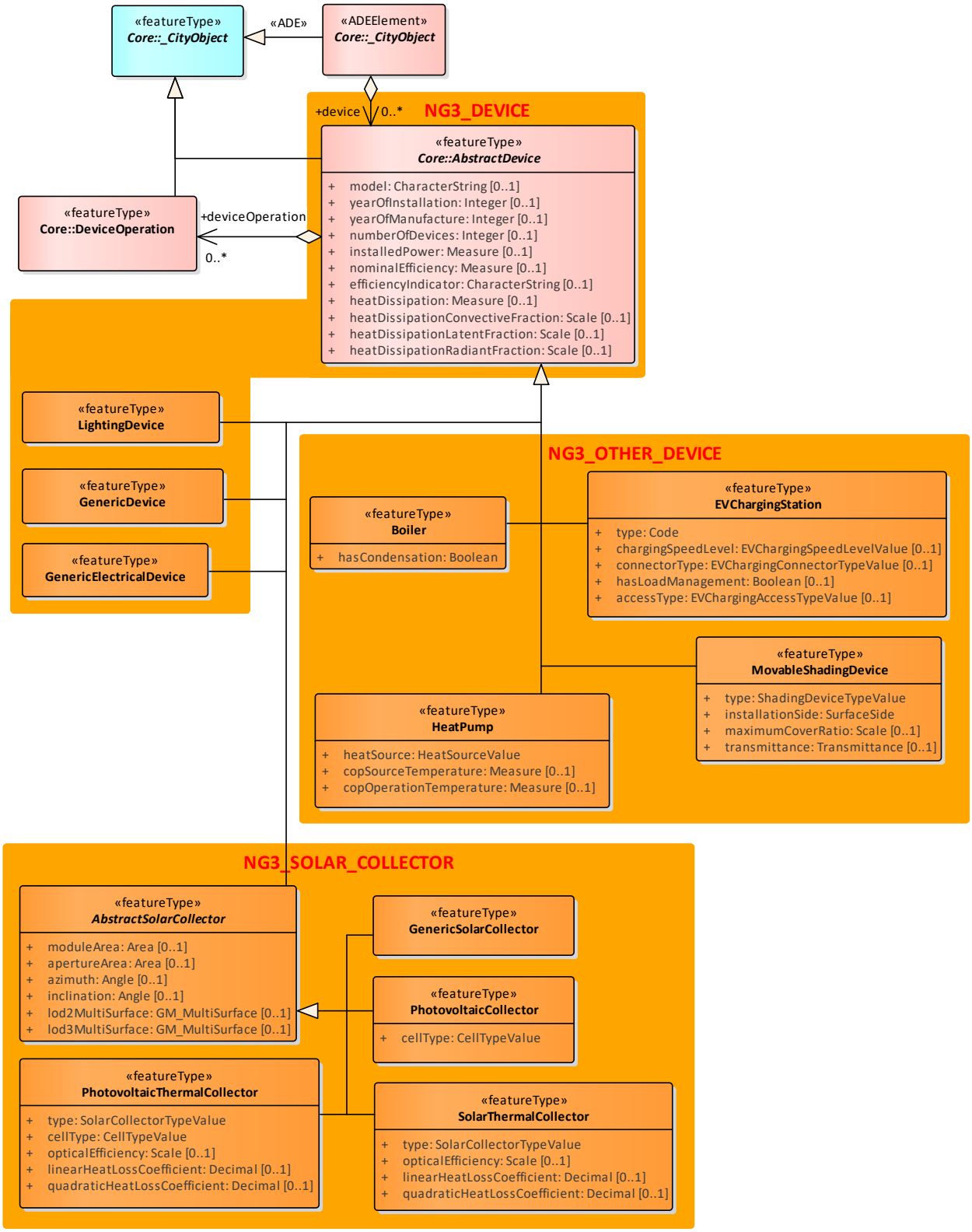
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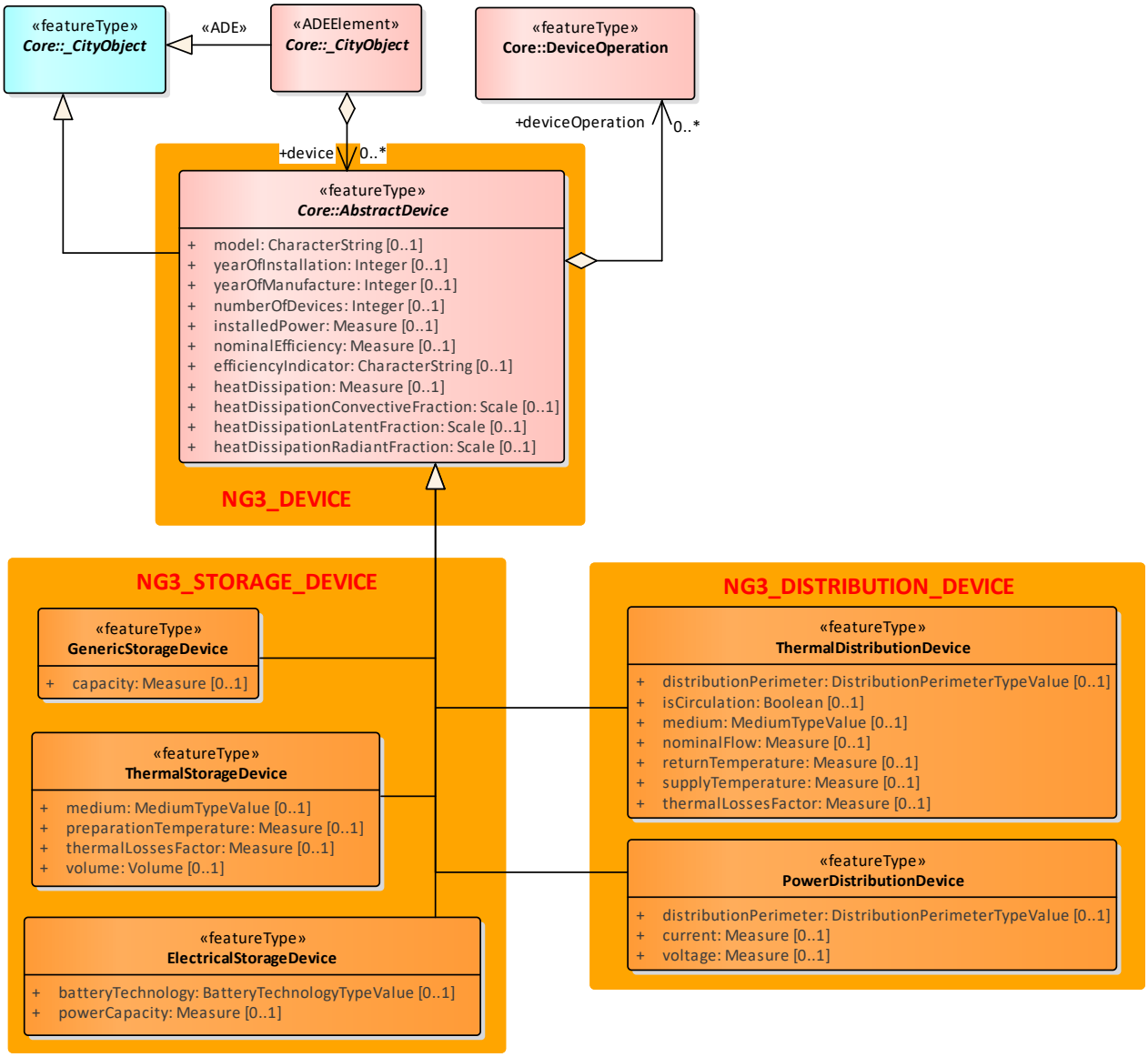
Building package part 3



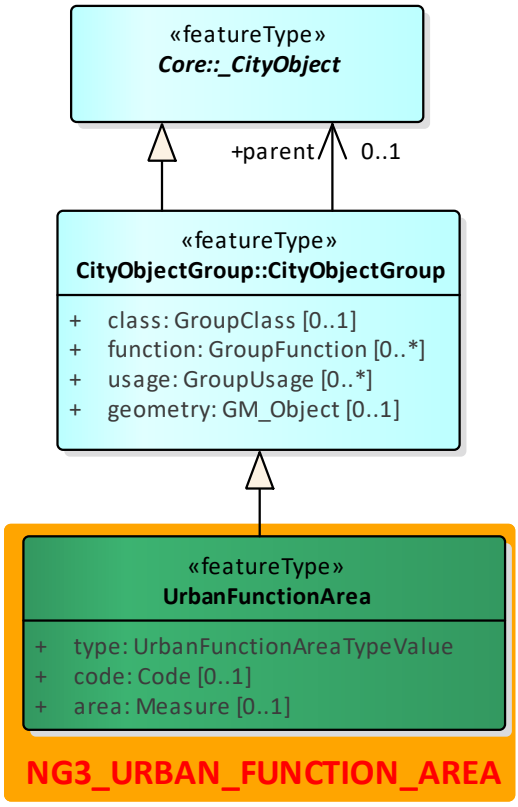
Devices package part 1



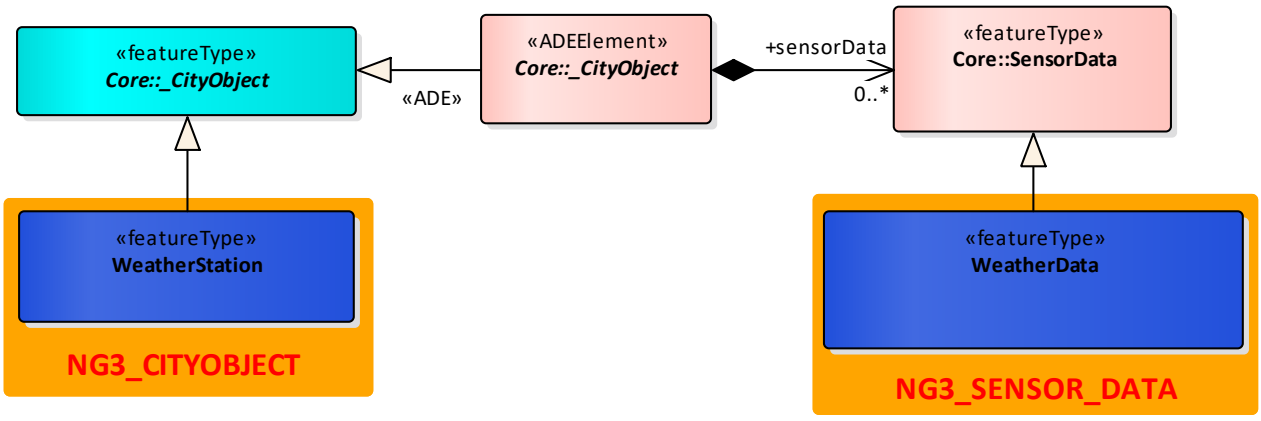
Devices package part 2



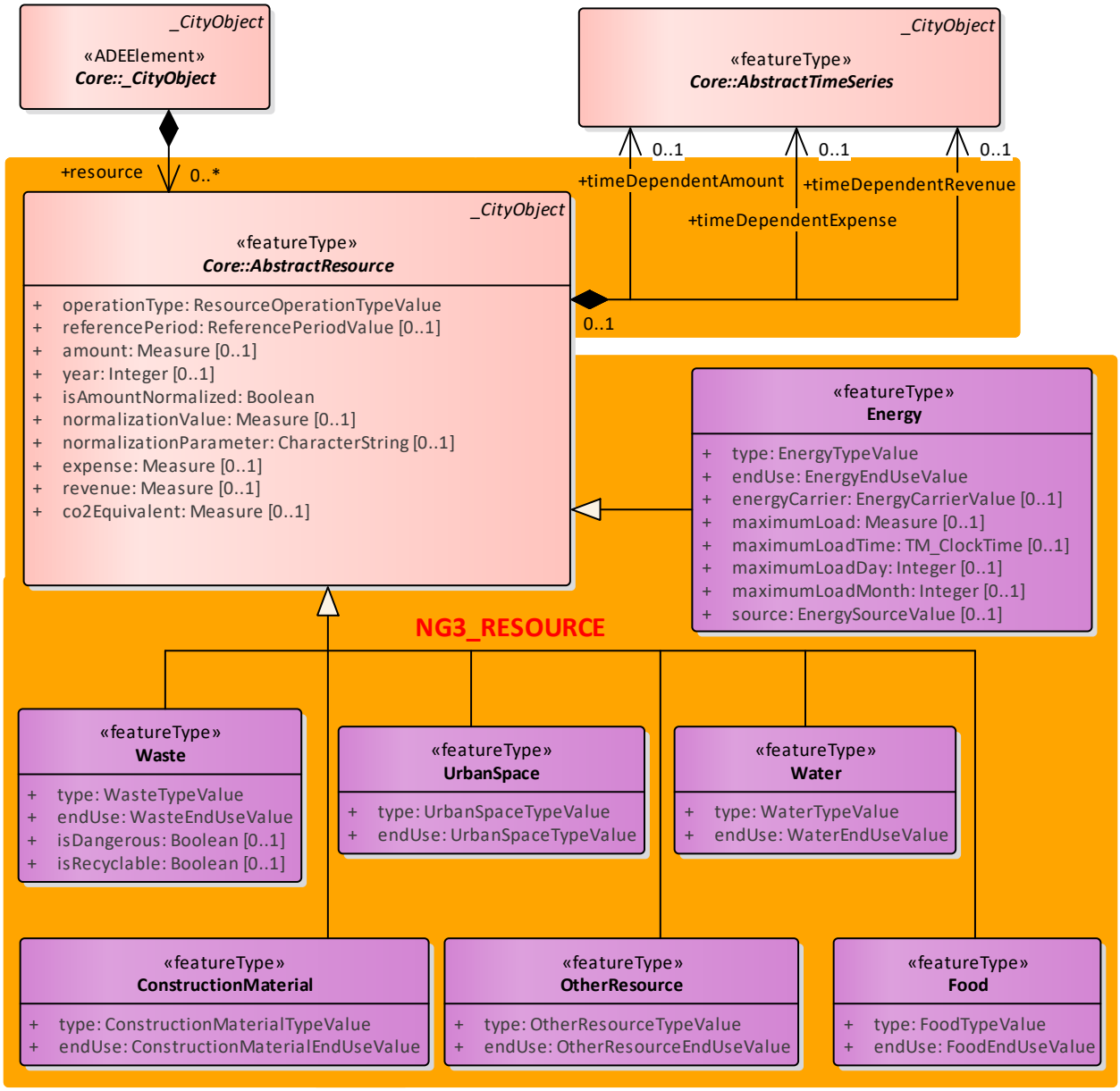
Urban function areas package



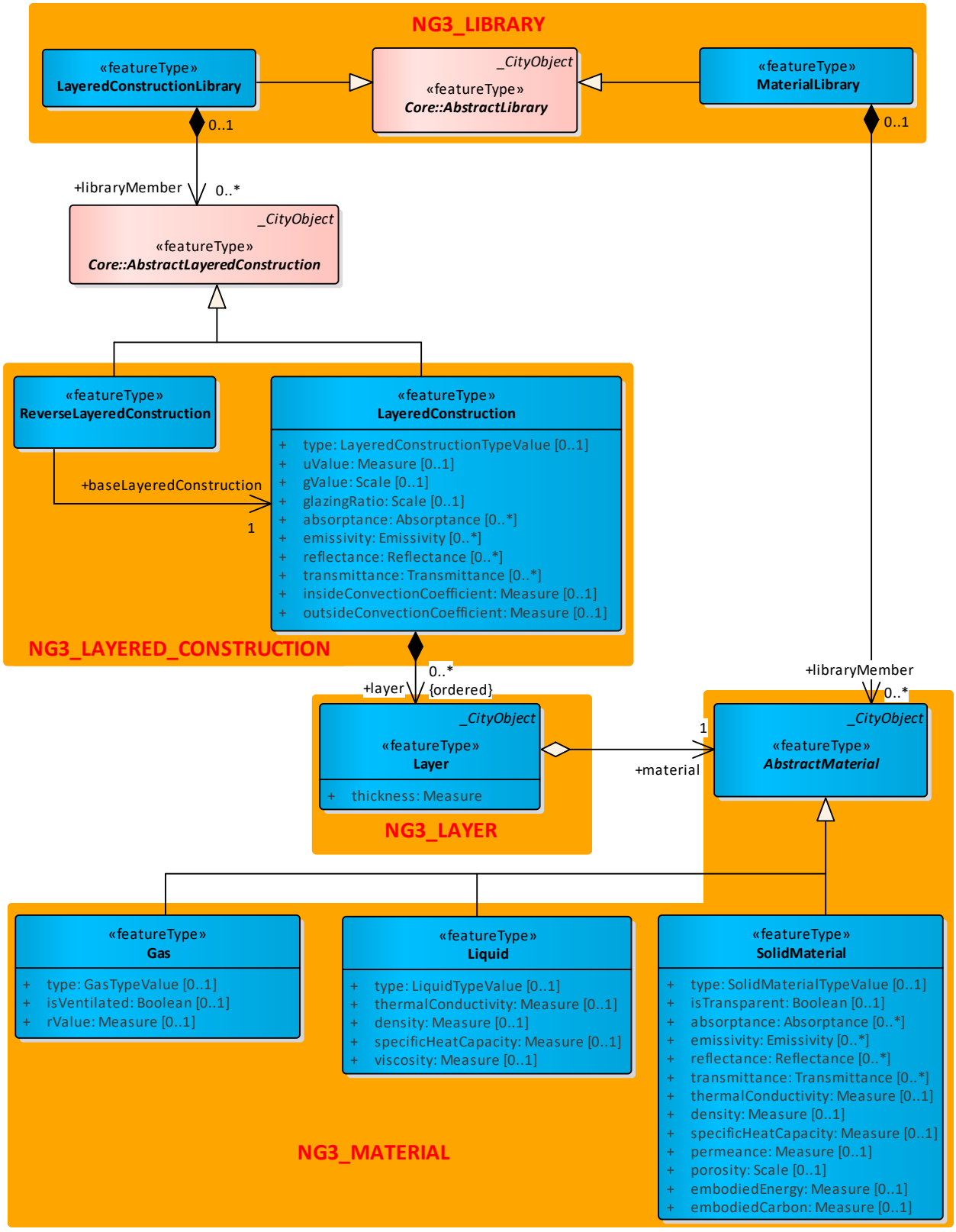
Weather station package



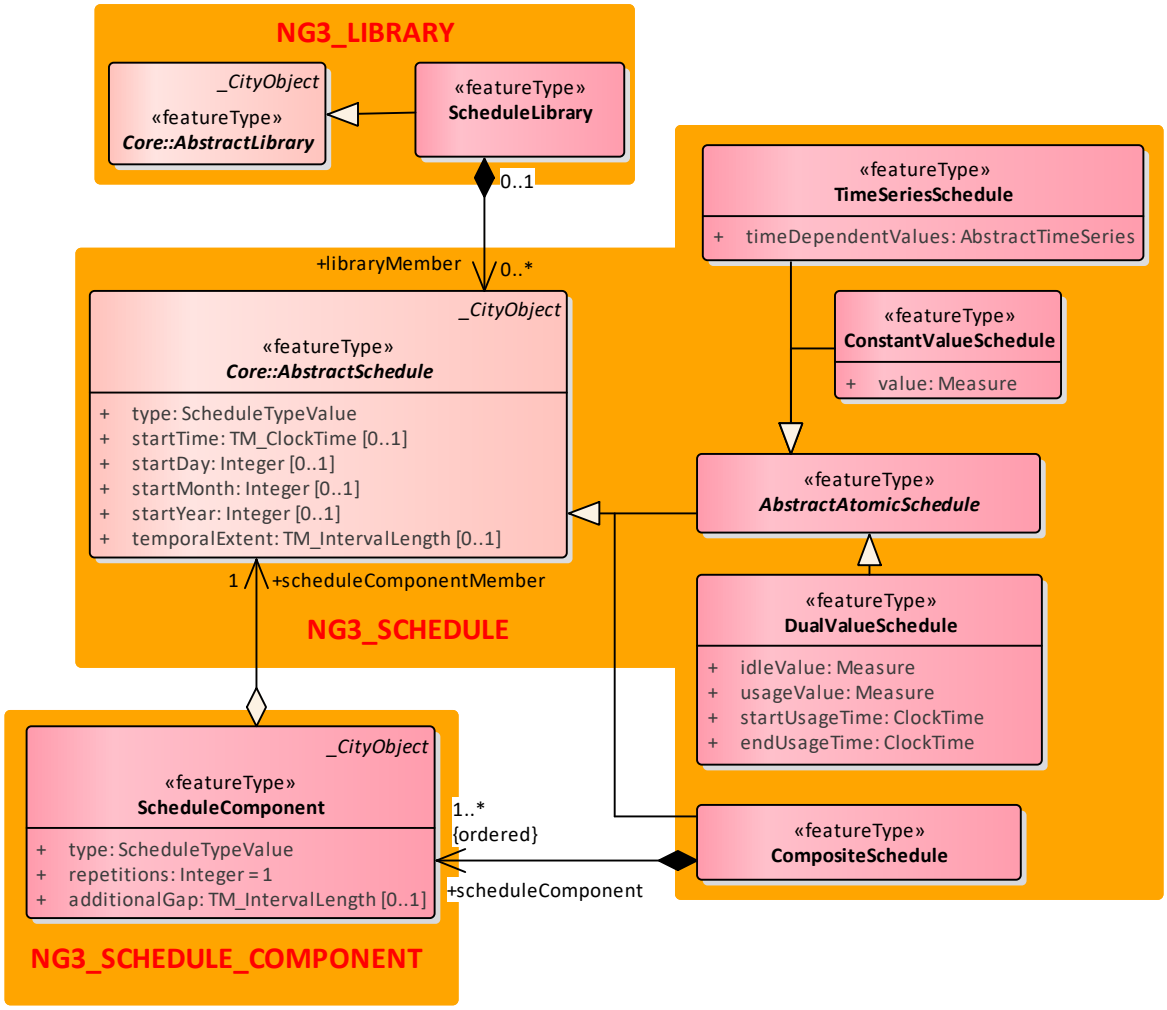
Resources package



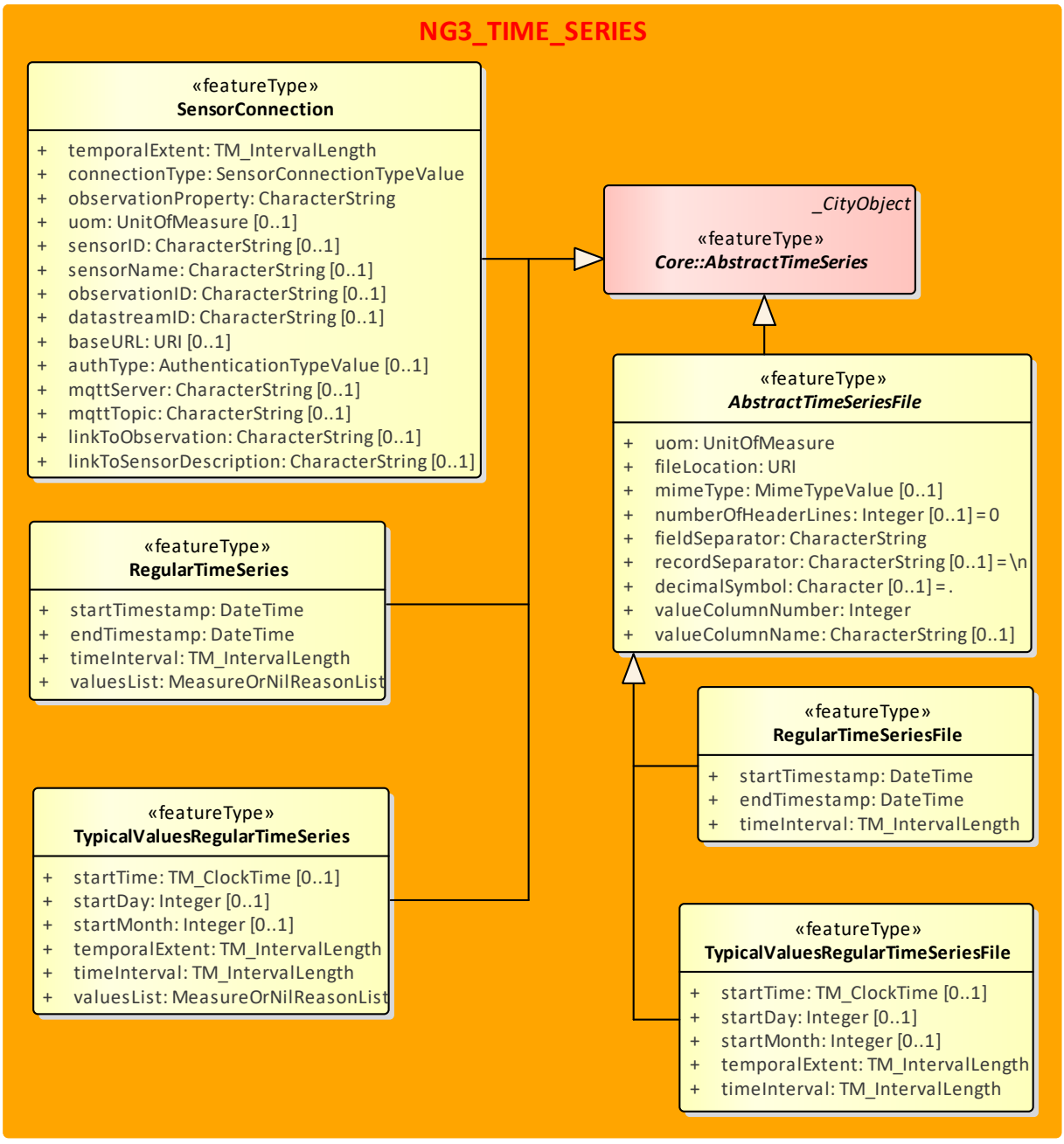
Layered construction package



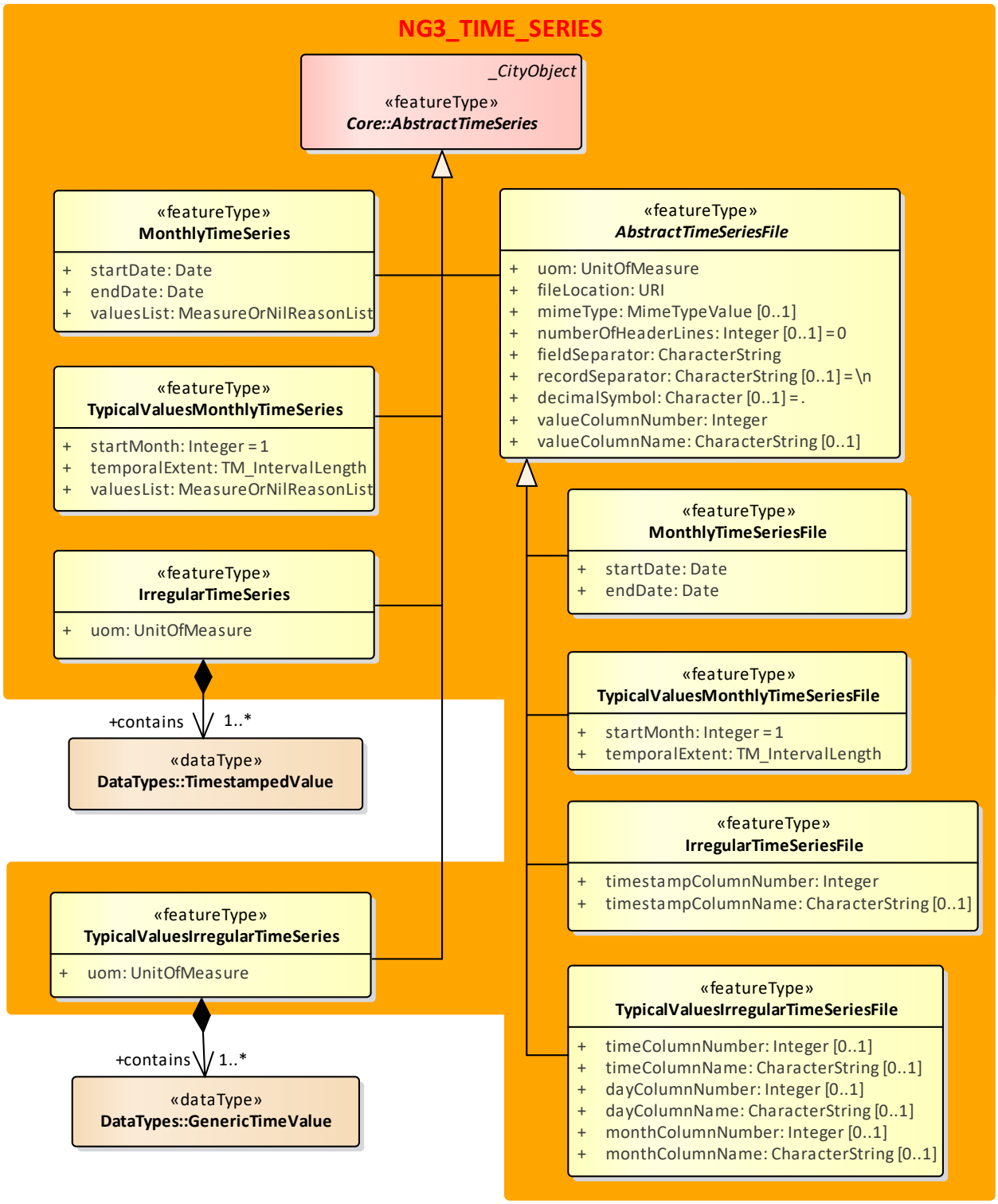
Schedules package



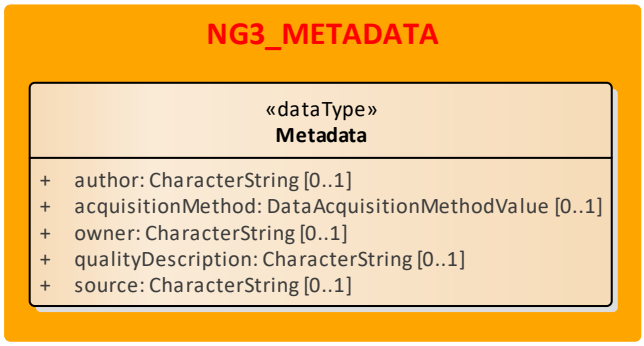
Time series package part 1

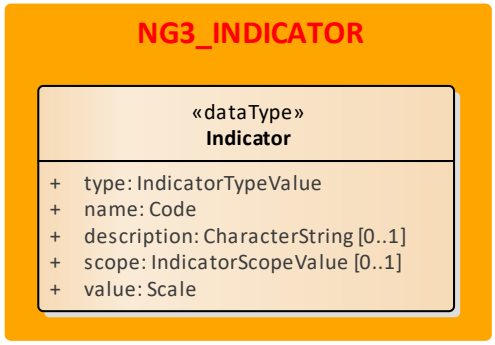


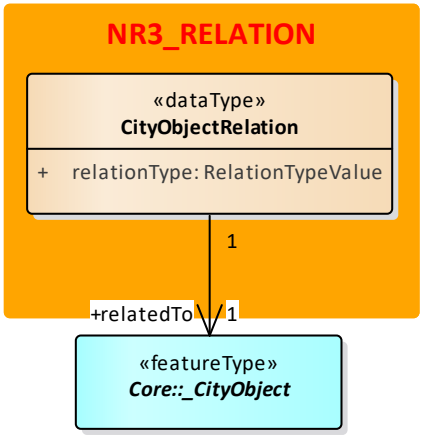
Time series package part 2

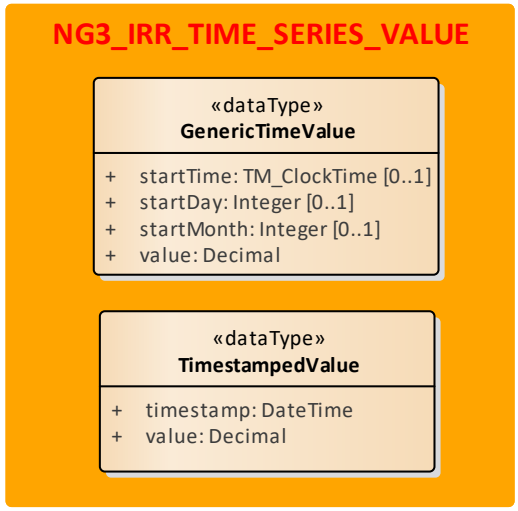


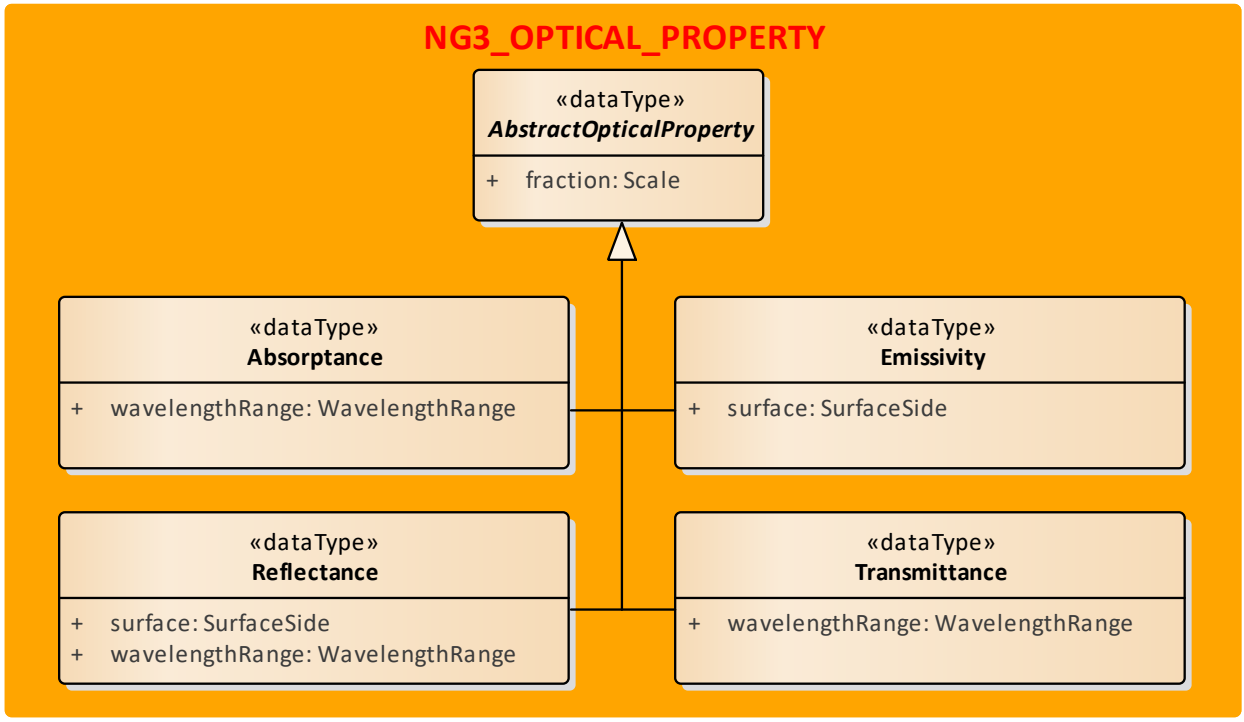
DataTypes package

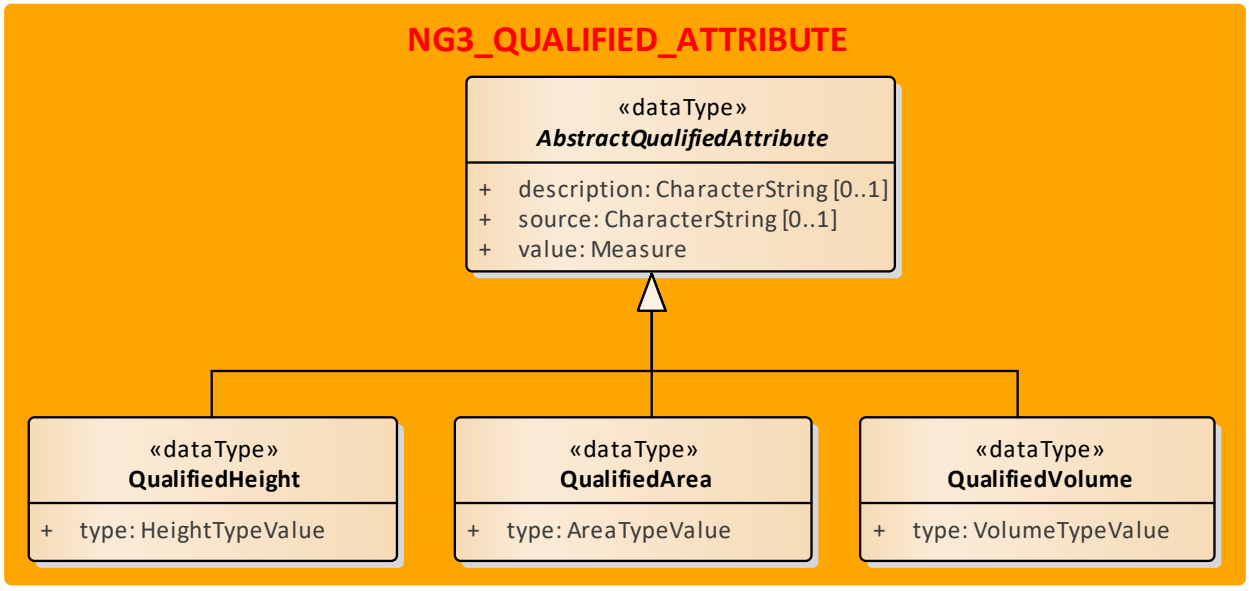









NG3_OPTICAL_PROPERTY
«dataType»
AbstractOpticalProperty
+ fraction: Scale
«dataType»
Absorptance
+ wavelengthRange: WavelengthRange
«dataType»
Emissivity
+ surface: SurfaceSide
«dataType»
Reflectance
+ surface: SurfaceSide
+ wavelengthRange: WavelengthRange
«dataType»
Transmittance
+ wavelengthRange: WavelengthRange


NG3_QUALIFIED_ATTRIBUTE
«dataType»
AbstractQualifiedAttribute
+ description: CharacterString [0..1]
+ source: CharacterString [0..1]
+ value: Measure
«dataType»
QualifiedHeight
+ type: HeightTypeValue
«dataType»
QualifiedArea
+ type: AreaTypeValue
«dataType»
QualifiedVolume
+ type: VolumeTypeValue

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